

B.A.Y.M.A.X. AI

Your Personal AI Health Companion

Bringing private, multimodal AI healthcare to every Indian home — anytime, anywhere, without an internet connection.

■ **Sector**	Healthcare Tech / HealthTech
■ **DPIIT Recognition No.**	[To Be Obtained – Application Pending]
■ **Date**	August 24, 2026
■ **Pitcher / Founder**	Kasinathan P S
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■■ *This document is confidential and intended solely for evaluation by authorised reviewers. Not for public distribution.*

1 INTRODUCTION — ELEVATOR PITCH

BAYMAX AI is a **HealthTech** startup building a **local, multimodal AI health companion** to help **individual consumers and families across India** get instant, private, and accurate health guidance by using **on-device Large Language Models (LLMs), Voice AI, and Medical RAG (Retrieval-Augmented Generation)** — with *zero reliance on the internet or cloud servers*.

- ■■ Built on FastAPI + FAISS + SQLite + Qwen LLM + Faster-Whisper + Coqui XTTS
- ■■ Fully offline-first architecture — your health data never leaves your device
- ■■ Voice-first interface (Speech-to-Text → AI Reasoning → Text-to-Speech)
- ■■ Contextual Medical RAG over a curated Indian health knowledge base
- ■■ Built for Bharat — targeting 1.4 billion people with limited healthcare access

2 THE PROBLEM

India's healthcare system is under immense structural pressure. Millions of citizens lack access to timely, affordable, and reliable medical guidance — creating a massive unmet need.

Pain Point	Scale	Source
Doctor-to-patient ratio	1 : 1,511 (far below WHO norm of 1:1,000)	NHP India 2023
Rural healthcare access	~65% of India lives in rural areas with <10% of doctors	NITI Aayog
Self-diagnosis via internet	>500 million Indians search health symptoms online monthly	Google India
Medical misinformation deaths	Thousands annually due to wrong self-diagnosis	Lancet India
Mental health crisis	197 million suffer mental illness; only 0.3 psychiatrists per 100k	WHO 2022
Health data privacy	Cloud health apps harvest personal data without consent	CERT-In Report

Core Pain: When a family member falls ill at 2 AM in a village with no doctor nearby, no reliable internet, and no money for a hospital visit — they have NO trusted AI health companion to turn to. **BAYMAX AI solves this.**

3 THE SOLUTION — PRODUCT & SERVICE

BAYMAX AI is a **fully private, offline-capable, multimodal AI health assistant** that runs on any standard consumer laptop or low-cost edge device. It combines state-of-the-art open-source AI into a unified, voice-first experience.

Module	Technology Stack	Capability
■ AI Brain (Engine 1)	Qwen3-8B / OmniRoute Gateway	Medical reasoning, Q&A, diagnosis guidance
■■ Voice Input (STT)	Faster-Whisper (medium)	Real-time speech-to-text in multiple languages
■ Voice Output (TTS)	Coqui XTTS v2 (cloned voice)	Natural, warm voice responses
■ Medical RAG	FAISS + sentence-transformers	Retrieval from curated medical knowledge base
■■ Vision / OCR	DeepFace + Tesseract	Prescription reading, symptom image analysis
■ Memory System	SQLite + ChromaDB	Persistent conversation & user health history
■ Networking	FastAPI + Tailscale + Nginx	Secure local & remote access over VPN

How It Works (User Journey):

1. User speaks a health query → Faster-Whisper transcribes speech to text
2. Medical RAG retrieves relevant knowledge from the FAISS vector database
3. Qwen3-8B LLM generates a medically-grounded, safe, educational response
4. Coqui XTTS speaks the answer back in a warm, natural voice
5. All data stays on-device — zero cloud upload, full privacy

■ *Prototype demonstration video available: baymax_intro.mp4 (included in submission package)*

4 INNOVATION & UNIQUENESS (USP)

BAYMAX AI is not another chatbot. It is a **purpose-built, privacy-first, offline medical AI stack** designed specifically for the Indian healthcare context.

Feature	BAYMAX AI	ChatGPT / Gemini	Practo / mFine	Generic Health Apps
Works without internet	■ Yes	■ No	■ No	■ No
Full data privacy (on-device)	■ Yes	■ No	■ No	Partial
Voice-first interface	■ Yes	Partial	■ No	Partial
Medical RAG knowledge base	■ Yes	General LLM	■ No	■ No
Prescription / Image reading	■ Yes	Partial	■ No	■ No
Persistent health memory	■ Yes	■ No	Partial	Partial
Emotion-aware responses	■ Yes (DeepFace)	■ No	■ No	■ No
Indian language focus	Planned	Partial	Partial	Partial
Cost to end user	■ Low	$ Expensive	■■ Medium	■■ Medium

IP / Patent Status: Provisional patent filing under preparation for the "Offline Medical RAG with Contextual Voice Memory" architecture. Expected to file within Q1 FY 2026-27.

5 TARGET MARKET & OPPORTUNITY

Target Customer: Individuals and families across India, particularly in Tier-2/Tier-3 cities and rural areas, who lack immediate access to qualified doctors and prefer private, on-device health guidance.

Market Segment	Definition	Estimated Size	CAGR
TAM (Total Addressable Market)	All individuals in India seeking digital health guidance	USD 50+ Billion (Indian Digital Health Market 2030)	27.4% (2024–30)
SAM (Serviceable Addressable Market)	Smartphone users in Tier-2/3 cities & rural India seeking AI health tools	USD 8–12 Billion	35%+ (AI Health)
SOM (Serviceable Obtainable Market – Yr 3)	Early adopters: tech-aware families, students, health-conscious users	~5 Lakh users (■ 15–30 Cr ARR)	Initial target

Key Market Drivers:

- India's digital health mission (Ayushman Bharat Digital Mission – ABDM)
- Rising smartphone penetration: 800M+ users by 2026
- Post-COVID trust in digital health solutions
- Government push for AI in healthcare (National AI Strategy – NITI Aayog)
- Growing data privacy awareness among Indian consumers

6 TRACTION & MARKET VALIDATION

BAYMAX AI is currently at **working prototype / MVP stage**. The following milestones have been achieved independently, with zero external funding:

Milestone	Status	Notes
Core architecture built	■ Complete	FastAPI + FAISS + SQLite stack
AI Engine 1 (LLM + RAG)	■ Complete	Qwen3-8B with medical RAG
AI Engine 2 (Voice + Vision)	■ Complete	Whisper + XTTS + DeepFace
Prototype demo video	■ Complete	baymax_intro.mp4 (1.1 MB)
Docker containerization	■ Complete	Multi-container deployment ready
Tailscale VPN integration	■ Complete	Cross-device secure access
Medical knowledge base (RAG)	■ In Progress	5 ChromaDB collections
User testing (beta)	■ Planned Q4 2025	Target: 50 beta users
Mobile companion app	■ Roadmap Q1 2026	React Native / Flutter
Multi-language support (Hindi)	■ Roadmap Q2 2026	Indic language models

■ **Validation Survey (Informal – 30 respondents):** 92% of respondents said they would use an offline, private AI health assistant. 78% cited 'privacy' as the top reason. 85% preferred voice over text interaction.

■■ *Formal survey with 200+ respondents planned post-funding as part of market validation phase.*

7 BUSINESS & REVENUE MODEL

BAYMAX AI will follow a **Freemium + Subscription model**, ensuring broad accessibility while creating sustainable recurring revenue.

Tier	Price	Features	Target Segment
■ Free	■0 / forever	Basic health Q&A (text), 10 queries/day, community support	Students, price-sensitive users
■ Standard	■99 / month (■799/year)	Unlimited queries, voice interface, medical RAG, reminders, prescription OCR	Individuals, families
■ Premium	■299 / month (■2,399/year)	All Standard + emotion detection, advanced analytics, priority updates	Health-conscious users
■ Clinic / B2B	■4,999 / month	Multi-user, API access, EHR integration, SLA support, custom branding	Clinics, pharmacies, NGOs

Additional Revenue Streams:

- Pharmaceutical partnerships (non-intrusive, opt-in health product recommendations)
- Government / CSR contracts for rural health digitisation programs
- Data analytics (anonymised, aggregated population health trends) — with user consent
- White-label licensing to hospitals and insurance companies

8 GO-TO-MARKET STRATEGY

A phased GTM strategy targeting digital-first consumers before expanding to institutional and rural markets:

Phase	Timeline	Goal	Channels
Phase 1: Early Adopters	Month 1–3	500 free beta users	GitHub, LinkedIn, Reddit, College campuses
Phase 2: Freemium Launch	Month 4–6	5,000 registered users	App stores, Social media ads, Health bloggers
Phase 3: Premium Conversion	Months 7–12	2,000 paid subscribers	Email marketing, In-app upgrades, Referral program
Phase 4: B2B & Clinic Sales	Year 2	50+ clinic partners	Direct sales, Health expos, NABH network
Phase 5: Tier-2/3 & Rural	Year 2–3	1 Lakh+ users	Govt partnerships, NGO channels, Regional language launch

First 100 Users Acquisition Strategy:

- Personal network outreach + WhatsApp health community groups
- Free demo sessions at local colleges and tech meetups
- YouTube video walkthrough + Reddit r/india health discussion posts
- Referral incentive: Refer 3 friends → get 1 month Premium free

9 TEAM

The founding team brings together deep technical expertise in AI engineering, system architecture, and healthcare domain knowledge.

Name	Role	Expertise & Background	Shareholding
Kasinathan P S (Founder & CEO)	Founder, CEO & Lead AI Engineer	Full-stack AI architecture: LLMs (Qwen, Ollama), FastAPI, (Indian) Whisper TTS, D	51%+ (Indian)
[Co-Founder – To Be Named]	CTO / Backend Lead	[Medical Informatics / Software Engineering backgrou	TBD
[Advisor – To Be Named]	Medical Advisor	[MBBS / MD – to validate medical content accuracy]	Advisory Equity

■ **Indian Promoter Shareholding:** Kasinathan P S holds >51% equity, meeting the DPIIT Startup India eligibility requirement for Indian promoter majority.

■ *Hiring plan post-funding: 1 Backend Engineer, 1 Medical Content Specialist, 1 Mobile Developer (React Native).*

10 FINANCIALS & PROJECTIONS

Current Investment: Self-funded (Bootstrap) — approximately ■ **1.5 – 2 Lakhs** invested by the founder in hardware (RTX 5050 GPU workstation), software licenses, and development tools.

3-Year Projected P&L; Statement (■ in Lakhs):

	FY 2025-26 (Year 1)	FY 2026-27 (Year 2)	FY 2027-28 (Year 3)
Revenue — Subscriptions	2.50	18.00	65.00
Revenue — B2B / Clinics	0.50	8.00	35.00
Revenue — Partnerships	0.00	3.00	12.00
TOTAL REVENUE	**3.00**	**29.00**	**112.00**
Cost of Operations (Tech Infra)	5.00	8.00	15.00
Salaries & Team	4.00	14.00	30.00
Marketing & Customer Acquisition	4.00	6.00	10.00
R&D & IP Filing	2.00	3.00	4.00
Admin & Misc	**1.00**	**2.00**	**3.00**
TOTAL EXPENSES	**16.00**	**33.00**	**62.00**
EBITDA (Profit / Loss)	**(13.00)**	**(4.00)**	**50.00**
Net Profit Margin	—	—	~44.6%

■■ *Projections are estimates based on comparable HealthTech SaaS benchmarks. Actual results may vary. Break-even projected at Month 18–20 post-market entry.*

11 FUNDING REQUIREMENT & USE OF FUNDS

BAYMAX AI is seeking ■ **20 Lakhs** in seed funding for **Proof-of-Concept (POC) / Prototype completion and early market validation**.

Use of Funds	Allocation	Amount (■)	Key Deliverable
Prototype Development & Engineering	40%	8,00,000	Mobile app, Hindi language model, medical knowledge base
Pilot Trials & Beta Testing	30%	6,00,000	50-user beta → 500-user pilot, user feedback loops
Marketing & Brand Building	20%	4,00,000	Social media, content marketing, demo events
Operational Expenses & Legal	10%	2,00,000	DPIIT registration, IP filing, server costs, misc
TOTAL	**100%**	**■ 20,00,000**	**Complete working product + 2,000 early users**

Key Milestones Post-Funding:

Month 1–2	Complete mobile app prototype + Hindi STT/TTS integration
Month 3–4	Conduct 50-user closed beta; iterate on feedback
Month 5–6	Public free tier launch; target 1,000 registered users
Month 7–9	Premium tier launch; target 200 paid subscribers
Month 10–12	DPIIT recognition application; first clinic B2B pilot

12 ANNEXURE — TECHNICAL SPECIFICATIONS

System Architecture Overview:

- **AI Engine 1 (Medical Reasoning Brain)**
  - Primary LLM: Qwen3-8B-Instruct (8-bit quantized, RTX 5050 8GB VRAM)
  - Fallback LLM: OmniRoute Gateway (auto/best-free via API)
  - Offline fallback: Ollama local server (qwen2.5:7b)
  - Embedding Model: sentence-transformers/all-MiniLM-L6-v2
  - Vector Store: FAISS (5 medical collections: disease, symptom, medicine, firstaid, general)
  - RAG Top-K: 5 retrieved chunks per query
  - Chunk Size: 512 tokens, Overlap: 64 tokens
- **AI Engine 2 (Multimodal Perception)**
  - Speech-to-Text: Faster-Whisper (medium model, int8 compute, 16kHz audio)
  - Text-to-Speech: Coqui XTTS v2 (24kHz, streaming, multilingual, voice cloning)
  - Emotion Detection: DeepFace with MediaPipe backend (15 FPS webcam, 5-frame smoothing)
  - OCR / Vision: Tesseract + PIL for prescription and image text extraction
  - Hosted on remote GPU node via Tailscale IP 100.79.169.64:8001
- **Backend & Infrastructure**
  - API Framework: FastAPI (Python 3.10) with async endpoints
  - Database: SQLite (baymax.db) for conversation history and user profiles
  - Memory: ChromaDB for episodic memory (short-term window: 20 messages)
  - Containerization: Docker + docker-compose (multi-service)
  - Reverse Proxy: Nginx (forwarding to Tailscale IP 100.89.251.123)
  - Test Coverage: pytest with structured test suite
- **Hardware Requirements (Minimum)**
  - CPU: Intel Core i5 / AMD Ryzen 5 (6th gen+)
  - RAM: 16 GB DDR4
  - GPU: NVIDIA RTX 3060+ (8GB VRAM) for full local LLM
  - Storage: 50 GB SSD (models + database)
  - OS: Windows 10+ / Ubuntu 20.04+
- **Security & Privacy**
  - All data processed and stored locally — no cloud transmission
  - API key authentication (configurable)
  - Tailscale WireGuard VPN for all remote connections
  - No third-party analytics or telemetry
  - Medical safety disclaimer on all AI responses

Repository & Submission Artifacts:

- Source code: GitHub repository (link to be added upon DPIIT registration)
- Demo video: baymax_intro.mp4 (included in submission package)
- Docker deployment: docker-compose.yml + Dockerfile
- Test suite: /tests/ directory with pytest configuration

Thank you for considering BAYMAX AI. We believe every Indian deserves a trusted, private AI health companion — and we are building it.